

The Copy Ceiling: An Input-Exposure Control for Ontology-Grounded Generation over Private Corpora

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Abstract

Enterprises increasingly need a language model that answers over a large, private, curated corpus the model never saw in pretraining; graph-based retrieval is the usual remedy, and it reports large “grounding uplift.” But when the gold answers are themselves derived from the injected corpus, that uplift conflates two behaviours: faithfully *delivering* exposed facts and *reasoning* over injected structure. We introduce the *copy ceiling*—a per-item, graph-derived input-exposure baseline: the recall a verbatim copy of the shown context already achieves—and the signed *gain over copy*, the first standing control that separates the two in graph-grounded generation. On a shipped ontology-serving node over a private corpus, unaided models score ≈ 0.26 and grounded ≈ 0.92 , yet across ten models from five providers the gain over copy is small and *uniformly negative* (-0.067 to -0.022): no model recovers gold beyond what its context already exposes. A production paired study (model held constant, only the serving path varied, cross-family judge) lifts judged quality by $+0.27$ pooled ($p = 0.004$) and $+0.79$ where curation is deepest; a four-arm negative-control decomposition binds that lift to scaffold *content*—the served block lifts quality ($+0.59$) while a verified seed-disjoint placebo does not ($+0.04$), the content-specific residual being $+0.35$. Out-of-domain the gated node does not regress: the general-set delta ($+0.05$, $[0.00, +0.13]$) lies inside a pre-registered ± 0.25 equivalence margin. We argue that grounding evaluations with corpus-derived gold should report a copy ceiling as standard, and that the ceiling plus a verified-inert placebo is a reusable protocol for telling exposure from reasoning before architecting.

1 Introduction

A recurring enterprise requirement is a language model that answers accurately over a large, private, domain-specific corpus—a customer’s product data, an organisation’s internal knowledge, a curated scientific ontology—that the model cannot have memorised because it was never in pretraining. Retrieval-augmented generation is the standard remedy for this gap between parametric memory and the knowledge an answer requires [11, 23], and it helps most where parametric memory is weakest: long-tail and proprietary facts [26]. Structured, graph-based retrieval sharpens it further [4, 7, 15, 29]. But evaluating such a system faces a measurement problem the headline “grounding uplift” hides: when the gold answers are themselves derived from the corpus that is injected, a large raw-to-grounded gain conflates two very different behaviours—faithfully *delivering* the exposed facts, and *reasoning* over the injected structure. Published knowledge-graph grounding reports the gain without a standing control that separates them.

We introduce the *copy ceiling*—a per-item, graph-derived input-exposure baseline—and the signed *gain over copy*, the first control that separates faithful delivery of injected facts from reasoning over ontological structure in graph-grounded generation. For each question the copy ceiling is the recall a no-op extractor of *exactly* the text the model was

shown would already score; the gain over copy is the model’s grounded recall minus that ceiling. It is the retrieval-augmented, per-item, continuous generalisation of the discrete input-only baselines long used to interpret reading comprehension [20] and natural-language inference [16, 32]. Its rhetorical role is a control the reader calibrates the headline number against, and the claim we draw from it is methodological: grounding evaluations whose gold is derived from the injected corpus should report a copy ceiling as standard [27, 36].

The setting makes the control load-bearing. Unaided, the models we test score ≈ 0.26 on in-domain questions—the domain is genuinely private, absent from parametric memory—and ≈ 0.92 given the curated context. Almost the whole of that uplift is exposure: across ten contemporary models from five providers the gain over copy is small and *uniformly negative* (-0.067 to -0.022), so no model recovers gold beyond what a copy of its context already exposes, and they differ only in how much surfaced gold they drop. This is not a defect to explain away. High-fidelity delivery of an authoritative, human-vetted source *is* the product for private-knowledge grounding: the answer is trustworthy because the source is, attributable to a named generation, and cheap because the expensive curation is amortised once rather than re-paid per query.

Contributions.

1. **The instrument (C1, singular).** The per-item copy ceiling and the signed gain over copy: a formal definition, an explicit assumptions list, and an interpretation that reads delivery fidelity, lossy restatement and reasoning-over-structure off a single signed quantity (§5).
2. **The evidence (C2).** Ten models are uniformly negative on gain over copy; a four-arm negative-control design (true / shuffled / masked / irrelevant scaffolds) attributes the gain to scaffold *content*; and a production-node paired study isolates the ecosystem feature by holding the model constant and varying only the serving path (§6; results in §7).
3. **The organising frame (C3).** A conjunctive, multivariate bar—in-domain delivery fidelity *and* out-of-domain non-regression, measured on two different instruments—within which the copy ceiling is the in-domain axis (§2, §9).
4. **The artefacts (C4).** An open harness (paired bootstrap, Wilcoxon, Cliff’s delta, Holm), the frozen question sets, all per-row data, and the served system itself.¹

2 The Private-Corpus Setting

The design decisions this paper measures are not preferences; they are forced by a chain of deployment requirements. We state the chain so that the evaluation bar—and the system in §4—read as constraints rather than choices. Each requirement carries one design consequence and its architectural decision record (ADR); Table 1 records where each is enforced. The object of study is a served node² over a corpus built by a separate, CI-gated pipeline³ and accelerated by an in-process HNSW index⁴.

2.1 The ecosystem in brief

This is a technical report on one component of a working stack (VisionFlow), and the component is easier to evaluate when the reader can see the whole data path. Knowledge enters at the corpus repository (*knowledgeGraph*): 8,138 Logseq markdown pages that compile losslessly into a pure-TBox OWL 2 ontology—8,146 classes, no individuals—published as versioned generations. Content arrives

¹The Loom serving node: <https://github.com/DreamLab-AI/loom>.

²The served node <https://github.com/DreamLab-AI/loom> (code under the repository licence; corpus terms per the sibling knowledgeGraph repository).

³The canonical builder <https://github.com/DreamLab-AI/knowledgeGraph> (published at <https://narrativegoldmine.com>) runs `pytest pipeline/tests` and `pipeline.validate` before publishing each generation.

⁴RuVector: <https://github.com/ruvnet/ruvector>.

there two ways: curated authorship under human direction, and, recently, automated extraction—output from OntoCast [14], an independently developed open-source ontology-extraction tool that is not our work, is staged through a preview-first import that reports identity collisions, refuses overwrites, and emits private review candidates rather than publishing machine output directly (the integration treats it as an upstream producer at a governed boundary; neither project vendors the other). A CI gate (consistency tests plus a native conflict detector) and a Whelk OWL 2 EL reasoner [5] classify each build and reject contradictions before publication; the reasoned closure (282,492 triples) ships with the generation. The *Loom* node measured in this paper mirrors a generation and serves it: lexical retrieval over class titles, confidence-gated injection of per-IRI markdown blocks, read-only SPARQL over the closure, and delegation to whatever local model sits behind its OpenAI-compatible façade (at the time of writing, Qwen3.8-27B on two workstation GPUs). *RuVector* supplies the vector substrate: the corpus embeddings live in its Postgres store, and an in-process HNSW index is wired as a semantic fallback—currently disabled by default because its measured recall (0.816) is below the pre-registered floor (0.87), a status we report rather than hide. Downstream, agent consumers (an email gateway, working agents) hold the façade URL and inherit grounding without integration work. Nothing in this description is load-bearing for the contribution; it exists so that §4–§7 can be read as measurements of a deployed system rather than a laboratory apparatus.

1. **Private-knowledge need $\Rightarrow$ curated, amortised corpus.** The model must answer over a corpus it cannot have memorised (raw ≈ 0.26 is the proof of privacy); the answer is grounded in a human-directed, provenance-tracked ontology whose expensive curation is paid once and reused per query. The copy ceiling is the evaluation-time expression of “the curation, not the model, carries the answer” (ADR-135 D2; ADR-136 D4).
2. **Trust $\Rightarrow$ human auditability at single-entity granularity.** A grounded answer is trustworthy only if a human can read and audit the served knowledge end-to-end; the canonical served unit is therefore one per-IRI markdown-with-ontology block—curated prose headed by its typed relations—and every retrieval port returns, or resolves to, an Iri addressing that unit (Invariant I-P1, “the prize”; RUST-ARCHITECTURE §0, ADR-136 D1).
3. **Auditability $\Rightarrow$ accelerators strictly behind the markdown.** Nothing a machine indexes may become the thing a human trusts instead of the markdown; the lexical index, HNSW, oxigraph SPARQL and the attestation ledger are projections that find, rank or attest the unit and resolve back to its IRI, enforced by an acyclic crate ring rather than reviewer vigilance (ADR-136 D1–D3; RUST-ARCHITECTURE §0–1).
4. **Privacy / LAN-confinement $\Rightarrow$ local delegation, no cloud on the hot path.** Content and answers must stay on the local network; generation is delegated to a LAN/local backend by URL and the augmentation hot path is fully in-process (ADR-135 D1; ADR-136 §3).
5. **Local-model churn $\Rightarrow$ model swap without consumer change.** Local models change often (Gemma $\rightarrow$ Muse-Glimmer $\rightarrow$ Qwen3.8-27B); a stable OpenAI-compatible façade makes the backend one configuration line and lets model identity ride in the result, never the endpoint—which is exactly what lets this study hold grounding constant while varying the model under test (ADR-135 D1).
6. **Bounded latency $\Rightarrow$ retrieval works with no model, fast.** Retrieval cannot depend on a model call: the scaffold, SPARQL and search endpoints need no model, and the lexical inverted index resolves in **2.02 ms** at the median over 8,146 class titles (ADR-136 §3).
7. **Injection interference $\Rightarrow$ confidence-gated, benchmark-first injection.** Injected context can displace correct parametric knowledge on off-ontology queries [25, 35, 38]; a single injection authority scales the budget to the retrieval score and skips below a threshold [26], and features that could add noise are benchmark-gated (ADR-136 D3).
8. **Never-mixed provenance $\Rightarrow$ attestation and boundary discipline.** Every grounded

Table 1: The private-corpus requirements arc: each business requirement, the design consequence it forces, and the point at which the consequence is enforced (not merely intended). ADR = the governing architectural decision record.

Requirement	Design consequence	Enforcement point
Private, unmemorised corpus	Human-curated ontology, curation amortised once	Sha-addressable generation; raw ≈ 0.26 (ADR-135 D2)
Human auditability	Per-IRI markdown-with-ontology as the served unit	Ports resolve to <code>Iri</code> $\rightarrow$ <code>CanonicalUnit</code> (I-P1)
Accelerators behind the unit	Indexes only find/rank/attest the markdown	Acyclic crate ring; adapter cannot mint the unit
LAN confinement	Model delegated by URL; in-process hot path	<code>DISTILL_BACKEND_URL</code> ; Profile A network-free
Model churn	Stable façade; model identity in the result	OpenAI-compatible /v1; one config line
Bounded latency	No-model retrieval path	Lexical index 2.02 ms p50; scaffold needs no model
Injection interference	Confidence-gated selective injection	Single injection authority; skip below threshold
Never-mixed provenance	Per-answer attestation; ontology-only boundary	<code>loom</code> block + <code>corpusNature</code> ; working graph never served

answer must trace to a named, versioned generation; each response carries a `loom` telemetry block and `corpusNature` honesty metadata, and the node serves only the published ontology, never the working graph (ADR-135 D6/D7; ADR-136 D5).

The bar these requirements imply is multivariate and conjunctive: excellent, attributable recall where the corpus is authoritative, *and* no regression on the general questions a model already answers. We introduce that framing here and reconcile the axes in §9.

3 Related Work

Knowledge-graph and private-corpus grounding. From the original RAG formulation [23] to graph-structured retrieval [7, 15], zero-shot knowledge-graph triple prompting [4], and the unifying roadmap and surveys [29, 31], the family grounds generation in retrieved graph content; GraphRAG motivates precisely our setting—answering over “private and/or previously unseen” collections [7]—and enterprise variants foreground the amortised-curation constraint [37]. Two features distinguish our object of study. Published work typically injects *instance* triples or opaque encodings—community summaries [7] or a GNN soft-prompt over a retrieved subgraph [17]—whereas we inject *schema-level* content (class definitions, typed relations, taxonomy) and keep the served unit a human-readable markdown block; and none of these reports an input-exposure control, so their uplift is uncontrolled for what the injected context already contains.

Baselines that reframe a result. A recurring genre installs a simple control and overturns the naive reading of a headline: a strong, simple RAG baseline matches elaborate pipelines under matched token budgets [36]; a controlled comparison reverses a prior “RAG wins” finding and adds a confidence router [24]; apparent argument comprehension is shown to be spurious-cue exploitation [28]. Our paper belongs to this genre: the copy ceiling is the control, and “uplift is exposure, not reasoning” is the reframing. The confidence router of Li et al. [24] and selective retrieval more broadly [3, 19, 26] are the design precedent for our injection gate—they route retrieve-versus-not; we gate inject-versus-abstain.

Input-only baselines and faithfulness metrics. The intellectual root of the copy ceiling is the input-only baseline: passage-only / question-only baselines reveal that much apparent competence is shortcut exploitation of what the input already exposes [20], as do hypothesis-only baselines in natural-language inference [16, 32]. Where these are discrete, per-task classifiers, the copy ceiling is per-item and continuous, for graph-grounded generation. On the metric side, faithfulness is separately measurable from correctness [1, 9, 40], and metric-first evaluations give the metric a formal definition with an explicit assumptions list [27] and a shortcut-robustness subsection [10]. FActScore’s precision-against-a-source and Adlakha’s token-overlap K-Precision are cousins of our exposure count; ALCE treats “copy the passage” as a shortcut to *reject*, whereas we make that copy the metric’s reference point. Contamination audits [13, 33] measure how much of a score is visible in the input post hoc; we make the deliberate exposure the reference a priori.

The nearest neighbour, and how we differ. The closest prior work establishes, by controlled removal and injection of the gold answer, that answer presence drives retrieval-augmented gains [2]; a parallel line defines an accuracy-given-gold-context upper bound to filter leakage out of benchmarks [34]. These prove the *phenomenon* and use gold-context as a benchmark-cleaning device. We differ in the artefact and its use: the copy ceiling is a per-item recall of a no-op extractor of *exactly* the shown text, reported as a standing control rather than a cleaning step, and the signed gain over copy is read *across models* as a faithfulness axis on which raw uplift cannot separate them. We keep the exposure and measure against it; they remove it. Judged arms use a cross-family judge to avoid self-preference [30, 39].

4 The Ontology Scaffold

The system measured here is the shipped serving node: a single static Rust binary organised as an eight-crate hexagonal workspace—a pure `loom-domain` core holding the port traits, five adapters, a `loom-scaffold` policy crate and a thin `axum` façade. The crate ring is the enforcement mechanism for the audit invariant of §2: only the domain crate mints a `CanonicalUnit`, so an adapter physically cannot return its own row, triple or vector as the served payload (Invariant I-P1). The node was adversarially audited by a different model family (gpt-5.4) with all findings remediated, and the Rust port is pinned byte-for-byte to the retired Python original by golden fixtures.

The canonical unit and the corpus. The served, reviewable unit is one per-IRI markdown-with-ontology block: curated research prose headed by its typed relations (`subclassOf`, `requires`, `enables`, `uses`, `relatedTo`, `contrastsWith`). The corpus is one sha-addressable *generation*—**8,146 concept classes** and **282,492 triples** in the Whelk-reasoned closure at the generation evaluated here—mirrored atomically so the node always serves a known, mixed-build-free generation; it is not baked into the binary. We explicitly reject the trajectory in which knowledge degrades into opaque community summaries or GNN-encoded subgraphs [7, 17]; the reviewable markdown is the moat, and the accelerators below only find, rank or attest it [12].

The lexical matcher and the confidence gate are the *sole* injection authority: every candidate flows through one policy that scales the scaffold budget to the retrieval score and injects nothing below a threshold—the mechanism by which the node avoids “going jagged” on off-ontology queries (§7.5). The graph store answers read-only, LIMIT-clamped SPARQL over the reasoned closure with no model call. An in-process HNSW semantic fallback over 384-dimensional embeddings is wired for out-of-vocabulary queries but ships *default-off*: its measured document-embedding recall (0.816) is below the 0.87 design floor, so it stays disabled pending a query-shaped embedding that clears a multivariate benchmark rather than a threshold relaxation—an honest red gate we return to in §10. Finally the model-swap seam: generation is delegated to whatever backend `DISTILL_BACKEND_URL` names (Qwen3.8-27B today), model identity rides in the response, and swapping the model changes no consumer—which is what lets §6 hold grounding constant and vary the model, or hold the model constant and vary the serving path.

Table 2: The request pipeline. Retrieval, gating and the scaffold serialisation need no model; only the final delegation calls one. The single injection authority (match + gate) is the sole path by which any candidate—lexical or, when enabled, semantic—reaches the model.

Stage	Mechanism	Model?
Match	Lexical inverted index over 8,146 class titles; 2.02 ms p50 $\rightarrow$ seed classes	no
Gate	Single injection authority: budget scaled to retrieval score, skip below threshold	no
Scaffold	Budget-clamped serialisation of the matched units’ markdown-with-ontology	no
SPARQL / search	Read-only, prologue-aware LIMIT-clamped oxigraph over the reasoned closure	no
Semantic fallback	In-process HNSW over 384-d embeddings; fires on lexical miss $\rightarrow$ candidate seeds into the same gate	<i>default-off</i>
Delegate	Inject scaffold into the last user message; call <code>DISTILL_BACKEND_URL</code>	yes

5 The Copy Ceiling

The copy ceiling is the paper’s primary contribution, so we give it a formal definition, an explicit assumptions list, and a robustness design of its own, following the metric-first precedent [10, 27].

5.1 Definition and assumptions

Fix a question q_i with gold target set G_i (the {slug, title} pairs the graph asserts as its answer), the exact scaffold text S_i shown to the model, and the model’s answer a_i . Let $m(t, X) \in \{0, 1\}$ be the deterministic surface matcher: $m(t, X) = 1$ when the normalised title of gold item t , or at least 80% of its length- ≥ 4 words, appears in text X . Define the exposed count and the per-item **copy ceiling**

$$e_i = \sum_{t \in G_i} m(t, S_i), \quad c_i = \frac{e_i}{|G_i|},$$

and the grounded recall $r_i = (\sum_{t \in G_i} m(t, a_i)) / |G_i|$. The copy ceiling is the recall of the no-op extractor that returns S_i verbatim as its answer: that answer scores exactly c_i by construction. For the raw arm S_i is empty, so $c_i \approx 0$.

Assumptions (when the instrument applies).

- (A1) **Deliberate exposure.** Questions, gold and scaffold derive from one graph, so the gold is present in the injected text by design. The ceiling *measures* this exposure; it is not a leak to remove.
- (A2) **Shared matcher.** c_i and r_i use the same matcher, so both are lower bounds under paraphrase, but their difference is first-order insensitive to the matcher’s under-counting; where paraphrase is a live threat we re-score with a semantic judge (§6).
- (A3) **Per-item, model-free ceiling.** c_i is a property of (q_i, S_i) only; for a fixed scaffold it is identical across models, which is what makes gain over copy comparable across a model sweep.
- (A4) **Graph-derived gold.** The ceiling is informative only when gold is derived from the injected corpus. Where gold is authored independently of the scaffold (the out-of-domain arm), $c_i \approx 0$ and a different instrument—judged quality—carries the axis.

5.2 Gain over copy

The signed **gain over copy** is $g_i = r_i - c_i$, with headline $\bar{g} = \bar{r} - \bar{c}$. It turns delivery fidelity into a signed, model-discriminating quantity and reads directly:

- $g \approx 0$ — *delivery fidelity*: the model restates the exposed facts and fabricates almost nothing. For private-knowledge grounding this is the target behaviour, not a shortfall.
- $g < 0$ — *lossy restatement*: the model drops surfaced gold, a faithfulness failure; the magnitude ranks models where near-identical raw uplift cannot.
- $g > 0$ — would evidence *reasoning over structure*: recovering gold the exposure check missed, for example by composing a relation the scaffold implies but does not state. We observe none in-domain; it is the signal a multi-hop arm would target.

A near-ceiling result therefore licenses “faithful delivery” and forbids the inference “reasoning over ontological structure”—the distinction the rest of the paper turns on.

5.3 Negative controls: what would move the gain

A single objection can sink the instrument: that a positive-looking delivery is lexical overlap between answer and scaffold, not knowledge transfer. We answer it with four arms that hold the model and the question fixed and perturb only the injected block (defined here; results in §7.4): *true* the block the node actually served, injected into the bare model. It should reproduce the live loom arm, proving the serving path adds nothing beyond the block itself.

shuffled the same block with its body sentences shuffled (wrapper kept, seeded). Same tokens, destroyed structure: if the gain survives, it was lexical soup.

masked the same block with every seed-class title replaced by **Entity- k** . Structure intact, entity names gone: if the gain survives, the score was matching names, not knowledge.

irrelevant the well-formed, on-corpus scaffold of a *different* question (a fixed derangement over the engaged items). If the gain survives, any ontology-shaped text would do.

The logic is differential. If gain over copy stays near zero under **shuffled**, **masked** and **irrelevant**, the ceiling is tracking exposure faithfully and the delivery is content-bound; if it moves, the ceiling was leaking through surface form or position. This is the experiment that turns the instrument from plausible to confirmed, and it is the direct analogue of a shortcut-robustness check for a faithfulness metric [10].

6 Method

Design. Both studies are paired within-model: every comparison is between two answers to the *same* question by the *same* model, so between-question difficulty variance cancels and the estimand is a paired per-question difference. The two studies isolate different variables—the ten-model sweep varies the *model* with grounding fixed; the new study varies the *serving path* with the model fixed.

Study 1 — the ten-model sweep (v1, summarised). Using the knowledge graph as its own oracle we template three question types from every sufficiently-defined class—**T-REL** (what a class requires / uses / depends on / enables; its parts), **T-TAX** (immediate and ancestral parents) and **T-COMMON** (a pair’s shared ancestor)—with gold the {slug, title} targets the graph asserts. Seed 42 yields **510 questions** over 15 domains (285 T-REL, 180 T-TAX, 45 T-COMMON). Each is asked *raw* and *scaffold* (a 1,500-token budget-clamped ontology extract prepended), across ten models from five providers under one deterministic configuration (**temperature**=0, **max_tokens**=2048; **reasoning_effort**=low only where thinking cannot be disabled, so a mandatory pass does not truncate the answer). Scoring is the lexical matcher of §5: a gold item is a hit when its normalised title, or $\geq 80\%$ of its long words, appears in the answer. This under-counts paraphrase, so absolute recall is a lower bound and we treat the *paired* delta—same question, scorer and model—as the signal, flagged honestly as a lexical estimand.

Study 2 — the production-node paired study (new). To isolate the ecosystem feature rather than the model, we hold the model constant (Qwen3.8-27B on the reference GPU host) and vary only the serving path. The *loom* arm calls the production Rust node’s `/v1/chat/completions`, which retrieves, confidence-gates and injects the scaffold and returns per-answer telemetry (engagement, injected tokens, fusion path) in the response `loom` block; the *raw* arm calls the same model backend directly with no scaffold. Both use `temperature=0` and `max_tokens=1536`, below which the reasoning backend truncates to empty. Questions are three frozen sets—*arcane* (deep in-domain), *thin* (sparsely-covered) and *general* (out-of-domain)—of **117** questions in total (24 arcane, 33 thin, 60 general), reused from the v1 sweep so the arms are judge-comparable. The four negative-control arms of §5.3 run against the same node: scaffolds are fetched from the LLM-free `/loom/scaffold` endpoint—so every control injects exactly the block the *loom* arm served—and presented to the bare model as a system message.

Cross-family semantic judge. Because Study 2 spans in-domain and out-of-domain questions and must survive the paraphrase objection, answers are scored not lexically but by a reference-guided **0–5 rubric** judge. The judge is a different model family from the candidate—an OpenRouter GPT judge grading the Qwen candidate, never the candidate’s own family—to avoid self-preference [30, 39]; the gold references are the graph-derived targets in-domain and frontier-authored key points out-of-domain.

Statistics. The headline per set and pooled is the paired mean difference with a seeded **10,000-resample bootstrap** 95% percentile interval [8, 21], complemented by a two-sided **Wilcoxon signed-rank** test and **Cliff’s delta** as a distribution-free effect size. Because the three sets are reported together, set-level *p*-values carry a **Holm–Bonferroni** correction [6]. Where the claim is *equivalence* rather than difference—the out-of-domain non-regression axis—we use two one-sided tests (TOST) against a pre-registered margin [22] of ± 0.25 , rather than reading a non-significant difference as no effect. In-domain, graph-derived questions cluster by class and domain [6]; following v1 we report a domain-clustered block bootstrap alongside the naive interval and an intention-to-treat delta that retains questions on which the gate did not engage [18].

7 Results

Reporting deviations (Study 2). Two departures from the pre-registered plan shape how the production-node numbers are read, and we state them up front. (i) *Budget-exhaustion re-runs.* At the `max_tokens=1536` budget, 42 of the 234 *loom/raw* completions (117 questions $\times$ two arms) spent the whole budget inside the reasoning trace and returned empty. Every affected pair was re-run at 4096 in *both* arms so the pairing is preserved; three questions were still empty at 4096 and are dropped, leaving the complete-case $n = 114$ paired analysis (the thin set falls from 33 to 30). (ii) *Controls not retried.* The four negative-control arms were not re-run at the larger budget, so at 1536 they carry substantial empty attrition—true 24/56 (42.9%), shuffled 20/56 (35.7%), masked 23/56 (41.1%), irrelevant 28/56 (50.0%)—and every control contrast is therefore complete-case over the surviving pairs, with the small, unequal n this implies. The attrition is not noise to discard but a finding in itself: at a fixed reasoning budget the empty rate rises with scaffold length and disorder (the live *loom* arm’s original-1536 empties were 2/24 arcane and 19/33 thin, against raw 6/24 and 15/33), a scaffold-length $\times$ reasoning-budget interaction we return to in §10. All control results below are read as lower-confidence, complete-case contrasts for this reason.

7.1 In-domain: delivery fidelity

On Gemini 3.7 Flash the paired design gives a large, tight uplift—raw recall 0.375, scaffold recall 0.942, paired delta **+0.567**, naive 95% CI $[+0.532, +0.603]$, domain-clustered 95% CI $[+0.524, +0.611]$ over $n = 510$ with zero errors and zero truncated responses (Figure 1). The raw

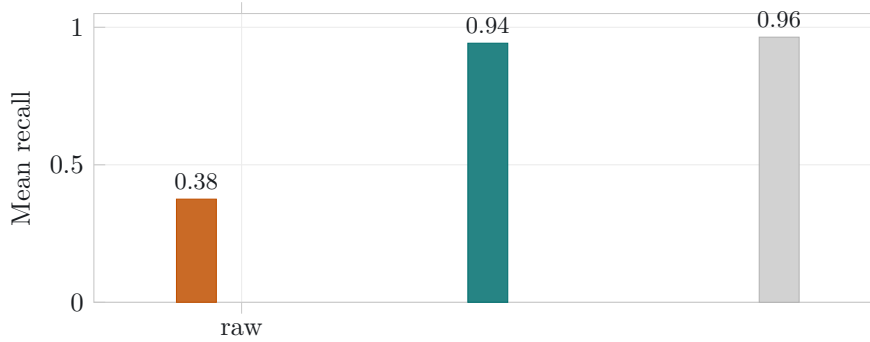


Figure 1: Gemini 3.7 Flash, $n = 510$. The scaffold arm (0.942) lands just under the *copy ceiling* (0.964)—the recall a no-op extractor of the injected context would already achieve. The $+0.57$ raw→scaffold uplift is almost entirely answer exposure, not reasoning over structure.

score confirms the domain is private—parametric memory answers barely a third. The copy ceiling then *characterises* the grounded score: the mean fraction of gold already present in the injected scaffold is **0.964**, and the model’s recall (0.942) tracks it within -0.022 . The system does exactly what a private-knowledge grounding layer should: the curated ontology carries the answers and the model delivers them back with high fidelity. The tiny negative gain says the model restates almost everything it is shown and fabricates almost nothing—faithful, attributable delivery, not free-association from parametric memory. Whether a model does more than restate—recovering gold the lexical check missed, or conversely *dropping* surfaced facts—is the axis §7.2 turns on.

7.2 Ten models: gain over copy

Table 3 runs the identical paired design across ten models from five providers under one deterministic configuration. Every model shows a large raw→scaffold uplift ($+0.57$ to $+0.78$) and every model lands just under the copy ceiling: scaffold recall spans 0.897–0.942 against a ceiling of **0.964** that is a property of the questions and the injected context—not the model—and is therefore identical across the row. The raw uplift is a near-constant restatement of exposed gold; it does not separate the models.

The signed **gain over copy** does. It is negative for all ten (Figure 2), from -0.067 (Gemini 2.5 Flash-Lite) to -0.022 (Gemini 3.7 Flash): on this lexical metric no model recovers gold *beyond* what the exposure check already finds, so they differ only in how much surfaced gold they *drop*. The ordering is interpretable and tracks recency and capability rather than provider—the newest, strongest models (Gemini 3.7 and 3.5 Flash, Claude Haiku 4.5) sit closest to the ceiling (most faithful restatement), while older or smaller models (Gemini 2.5 Flash-Lite, GPT-4.1-mini, Mistral-Small-24B) lose measurably more. The rank order of gain tracks that of raw capability, not scaffold recall (which is flat, 0.897–0.942): the instrument is reading a model property the headline uplift hides. Two models carried transport-level truncations under the fixed 2048-token budget (GLM-4.6: 174 truncated, 5 retried; Gemini 2.5 Flash-Lite: 16 truncated), recorded per row rather than hidden.

7.3 The production-node paired study

The ten-model sweep varies the model with a fixed offline scaffold; the deployment question is whether the *shipped* node—live retrieval, confidence gate, injection authority and all—moves answer quality when the model is held constant. Study 2 answers it: Qwen3.8-27B on the reference host, identical decoding, the only difference being whether the request passes through the Loom node (*loom*) or hits the backend directly (*raw*), scored 0–5 by the cross-family GPT judge of §6. Table 4 reports the paired result per set and pooled.

Pooled over $n = 114$, the loom path lifts judged quality by **+0.27** points ($[+0.11, +0.45]$, Wilcoxon $p = 0.0036$, Cliff’s $\delta = 0.14$), winning 24 pairs, losing 8 and tying 82. The effect

Table 3: Ten models, one deterministic configuration, $n = 510$ (seed 42), sorted by gain over copy (scaffold recall – copy ceiling). The copy ceiling (0.964) is a property of the questions and the injected context and is identical across models. The interval shown is the domain-clustered block bootstrap (the wider, primary one).

Model	Raw	Scaffold	Uplift [clustered 95% CI]	Ceiling	Gain
Gemini 3.7 Flash	0.375	0.942	+0.567 [+0.524, +0.611]	0.964	−0.022
Gemini 3.5 Flash-Lite	0.323	0.938	+0.615 [+0.565, +0.667]	0.964	−0.026
Claude Haiku 4.5	0.151	0.934	+0.783 [+0.741, +0.827]	0.964	−0.031
GLM-4.6	0.243	0.928	+0.685 [+0.641, +0.727]	0.964	−0.036
DeepSeek-Chat	0.347	0.924	+0.578 [+0.532, +0.622]	0.964	−0.040
Llama-3.3-70B	0.227	0.916	+0.688 [+0.632, +0.738]	0.964	−0.049
Qwen-2.5-72B	0.309	0.914	+0.605 [+0.562, +0.649]	0.964	−0.050
Mistral-Small-24B	0.285	0.906	+0.621 [+0.558, +0.682]	0.964	−0.058
GPT-4.1-mini	0.162	0.905	+0.743 [+0.697, +0.787]	0.964	−0.060
Gemini 2.5 Flash-Lite	0.227	0.897	+0.670 [+0.623, +0.716]	0.964	−0.067

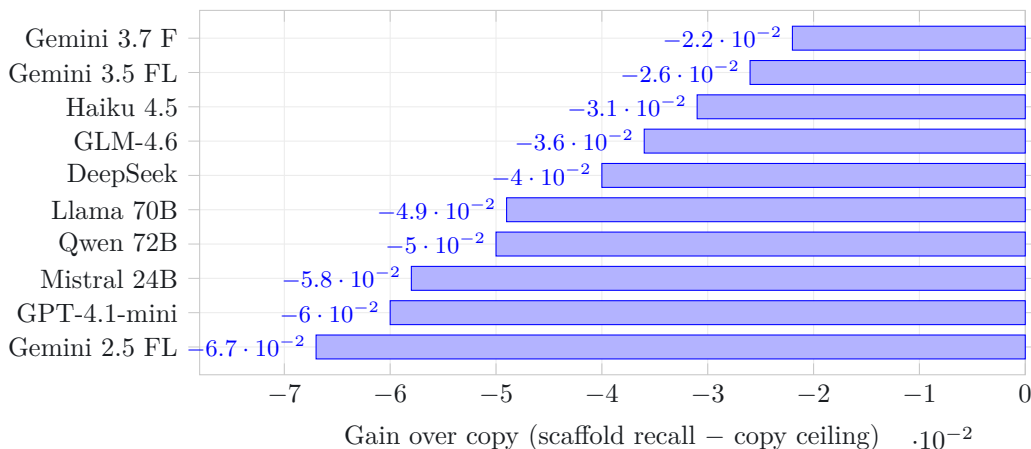


Figure 2: Signed gain over copy for all ten models—negative throughout. No model exceeds the copy ceiling on the lexical metric; the spread (−0.067 to −0.022) is the axis that separates them where the near-identical raw uplift does not. Models nearest zero (top) restate the vetted facts most faithfully.

concentrates where the corpus is deep: on the *arcane* set the lift is +**0.79** ([+0.17, +1.42], $p = 0.033$, Holm-adjusted $p = 0.099$, $\delta = 0.29$); on the sparsely-covered *thin* set it is a positive but non-significant +0.30 (−0.03, +0.63]; and on the out-of-domain *general* set it is a negligible +0.05 whose interval [0.00, +0.13] sits entirely inside the pre-registered ± 0.25 equivalence margin—non-regression established by CI-inclusion, not merely by a non-significant difference (the TOST reading; §7.5). The gradient—large where curation is deep, vanishing where the corpus is thin or the question is off-domain—is exactly the signature a genuine grounding effect should show, and the opposite of what a uniform prompt-formatting artefact would produce.

The telemetry corroborates that the gate behaves as designed. Injection engagement is 95.8% on *arcane*, 100% on *thin*, and drops to 78.3% on *general*—the gate declining to inject on roughly a fifth of off-domain questions, the mechanism behind the *general*-set null. Median injected context is $\approx 1,200$ –1,300 tokens across sets. Latency is at parity on the sets that matter for the claim: *arcane* medians 89.8s (loom) versus 93.4s (raw), *general* 66.0 versus 65.7s; the *thin* set alone runs slower under loom (172.4 versus 145.2s), consistent with its full engagement and longer scaffolds, and we report it rather than average it away.

7.4 Negative controls: attributing the gain

The live lift and the ten-model delivery both invite one objection: that the gain is lexical overlap between answer and injected block, not knowledge transfer. The four control arms of §5.3 test it

Table 4: Study 2, the production-node paired study: Qwen3.8-27B, model held constant, loom versus raw serving path, judged 0–5 by a cross-family GPT judge. Paired mean difference with seeded 10,000-resample bootstrap 95% CI, two-sided Wilcoxon p (Holm-adjusted across the three sets), Cliff’s δ , and win/loss/tie counts. Complete-case after budget-exhaustion re-runs (§7).

Set	n	Δ (loom–raw)	95% CI	p (Holm)	δ	W/L/T
arcane	24	+0.79	[+0.17, +1.42]	0.033 (0.099)	0.29	10/3/11
thin	30	+0.30	[−0.03, +0.63]	0.118 (0.237)	0.23	12/5/13
general	60	+0.05	[0.00, +0.13]	0.180 (0.237)	0.03	2/0/58
pooled	114	+0.27	[+0.11, +0.45]	0.0036	0.14	24/8/82

directly, each holding the model and question fixed and perturbing only the injected scaffold, fetched verbatim from the LLM-free `/loom/scaffold` endpoint. Table 5 reports the pooled (arcane + thin) contrasts; we read them as lower-confidence, complete-case comparisons for the attrition reasons of §7.

Three results triangulate the attribution. First, a **sanity check**: serving the exact block through the bare model (*true*) reproduces the live *loom* arm—loom – true is -0.06 $[-0.38, +0.25]$, indistinguishable from zero, confirming the serving path adds nothing beyond the block it injects. Second, the **content effect is real**: true – raw is $+0.59$ $[+0.16, +1.06]$ ($p = 0.026$)—injecting the served block lifts judged quality well above the bare model. Third, and decisively, the **placebo collapses**: an equally well-formed, on-corpus scaffold drawn from a *different* question (*irrelevant*) moves nothing—irrelevant – raw is $+0.04$ $[-0.54, +0.57]$ ($p = 0.73$). Subtracting the placebo from the content arm isolates the content-specific component, true – irrelevant = $+0.35$ $[0.00, +0.78]$ ($p = 0.091$): the majority of the raw-to-served lift is bound to *which* block is injected, not to the mere presence of ontology-shaped text.

The placebo had to be verified inert. This conclusion depends entirely on the irrelevant arm being a true placebo, and our first attempt was not. Constructing the “different question” donor by a shift-by-one derangement over the engaged items left donors in the *same* domain cluster as their targets—so the “irrelevant” scaffold routinely shared vocabulary and neighbouring concepts with the gold, and behaved like a weak content arm rather than a placebo. We rebuilt the derangement under a seed-IRI-disjointness constraint (every donor scaffold verified to share no seed IRI, and thus no seed class, with its target’s gold) and only then did the arm collapse to ≈ 0 . A placebo that is assumed inert rather than verified inert silently inflates the very baseline it is meant to control; the fix was cheap and the lesson is general.

Masking and shuffling lose comparatively little. Replacing every seed-class *name* with **Entity- k** (*masked*) still beats raw by $+0.42$ (ns, wide CI), and shuffling the block’s sentences (*shuffled*) by $+0.25$ (ns): the graded facts, not the surface names or their order, carry most of the signal—consistent with a judge grading factual content over lexical form. Finally, the effect is a **curation-depth gradient**: per set, true – raw is $+0.95$ $[+0.37, +1.63]$ ($p = 0.018$) on the deep arcane set but only $+0.08$ $[-0.46, +0.62]$ (ns) on the thin set. Where the corpus is rich the served block transfers real knowledge; where it is sparse there is little to transfer—the same gradient the live study shows, recovered independently through the controls.

7.5 Out-of-domain: the gate holds

Two instruments agree that grounding does not cost anything on the general questions a model already answers. The v1 judged arm graded a 60-question out-of-domain set 0–5 against frontier gold with a cross-family judge across five models: every paired harness–bare delta’s 95% interval includes zero—all questions -0.05 $[-0.11, 0.00]$; off-domain -0.09 $[-0.25, +0.02]$; and even the pre-specified worst case, off-domain questions on which the gate mis-fired and injected anyway, -0.23 $[-0.63, +0.05]$, still spanning zero (Table 6). The mechanism is the gate: when it correctly

Table 5: Negative-control contrasts, pooled arcane + thin, judged 0–5, paired mean difference with seeded bootstrap 95% CI and two-sided Wilcoxon p . Complete-case over surviving pairs; the control arms were not retried at the larger budget, so n varies by arm (§7). Read top to bottom: the serving path adds nothing (loom–true ≈ 0); the served block lifts quality (true–raw); the same-shaped placebo does not (irrelevant–raw ≈ 0); the content-specific residual is true–irrelevant.

Contrast	n	Δ	95% CI	Wilcoxon p
loom – true	32	−0.06	[−0.38, +0.25]	0.81
true – raw	32	+0.59	[+0.16, +1.06]	0.026
masked – raw	33	+0.42	[−0.09, +0.94]	0.15
shuffled – raw	36	+0.25	[−0.39, +0.89]	0.42
irrelevant – raw	28	+0.04	[−0.54, +0.57]	0.73
true – irrelevant	23	+0.35	[0.00, +0.78]	0.091

Per set (true – raw): arcane +0.95 [+0.37, +1.63], $p = 0.018$; thin +0.08 [−0.46, +0.62], ns.

Table 6: Out-of-domain non-regression (v1 judged arm): paired harness–bare quality delta (judge 0–5 vs frontier gold), five models, seeded paired bootstrap 95% CI. Every interval includes zero. The worst case is off-domain questions where the confidence gate mis-fired and injected anyway; the last row is where it correctly skipped.

Subset	Mean Δ	95% CI	n pairs
All general questions	−0.05	[−0.11, 0.00]	300
adjacent	+0.01	[−0.02, 0.05]	100
in-domain-general	−0.06	[−0.14, 0.00]	100
off-domain	−0.09	[−0.25, 0.02]	100
Off-domain, gate mis-fired (worst)	−0.23	[−0.63, 0.05]	40
Off-domain, gate skipped (correct)	+0.00	—	60

judged a query off-domain and skipped (60 of 100 off-domain gradings) the harnessed answer was identical to bare; the only degradation appeared when it over-fired, and it concentrated on the weakest model. The production study adds the equivalence reading the v1 arm lacked: the general-set $\Delta = +0.05$ [0.00, +0.13] lies wholly inside the pre-registered ± 0.25 TOST margin, so non-regression is established by CI-inclusion rather than inferred from a non-significant difference. As in v1 we report “no jaggedness detected” rather than “zero effect”—the intervals are wide and bound the regression near zero, they do not prove its absence.

8 Analysis: Exposure, not Reasoning

The results across both studies say one thing in several registers: the uplift a graph-grounding layer delivers over a private corpus is overwhelmingly *exposure*—faithful delivery of curated facts the model was shown—and not *reasoning* over injected structure. This section reads that claim off the instrument, then draws the two engineering consequences that reframe how such a system should be built and evaluated.

The headline is delivery fidelity, and the instrument is what reveals it. The ten-model sweep is the cleanest statement: raw uplift is huge and near-constant (+0.57 to +0.78), yet gain over copy is small and *uniformly negative* (−0.067 to −0.022). Read against a bare-model baseline every model looks transformed; read against the copy ceiling none exceeds what a verbatim copy of its own context already exposes, and they separate only by how much surfaced gold they drop. The number the field would report as the achievement (the uplift) is the number that carries the least information about the model; the number that discriminates (gain over copy) is invisible without the ceiling. That inversion is the whole case for reporting the ceiling as standard.

The live decomposition confirms it is content, not form. The production controls take the same conclusion from headline to mechanism. The served block lifts judged quality (true – raw +0.59); a same-shaped, on-corpus but wrong block does not (irrelevant – raw +0.04); the content-specific residual is +0.35; and masking names (+0.42) or shuffling sentences (+0.25) costs comparatively little. The facts carry the effect; the entity names and their order do not. Three caveats keep this honest: the arms are complete-case over unequal survivor sets, the contrasts do not sum arithmetically (different survivors per arm), and several are non-significant with wide intervals. But the qualitative pattern—content-specific lift present, verified-inert placebo absent—is exactly what “exposure, faithfully delivered” predicts and what “any ontology-shaped text primes the model” does not. The placebo episode is the load-bearing methodological point: the effect only appeared clean once the irrelevant donor was *verified* seed-IRI-disjoint; a placebo assumed inert had leaked domain vocabulary and manufactured a spurious baseline.

Consequence 1: the copy ceiling is a regime detector—compute it before you architect. The ceiling is not only a post-hoc control; it is a design-time diagnostic. Compute it on a representative question set *before* choosing an architecture. A near-1 ceiling (our setting: 0.964) means the corpus already contains the answer verbatim—this is a *serving regime*, where verbatim retrieval wins and generative re-encoding can only lose fidelity, so the engineering target is faithful delivery and the metric is gain over copy. A low ceiling means the answer is *not* present as-is and must be composed across retrieved pieces—a *synthesis regime*, where GraphRAG-style machinery (community summaries, multi-hop traversal, GNN soft-prompts) can earn its keep. The essential discipline is that in a synthesis regime that machinery must be measured *above the ceiling*, not above a bare-model baseline: an uplift over the bare model in a high-ceiling setting is delivery masquerading as reasoning. Much reported graph-RAG gain is, we suspect, un-diagnosed serving-regime exposure; the ceiling is the cheap test that tells the two apart before a team builds the expensive apparatus.

Consequence 2: value concentrates at curation, and the instrument doubles as a promotion gate. If the model contributes delivery and the corpus contributes the answer, then the place to invest is upstream, at curation time—which is where the economics of a private corpus already put it, since the curation is amortised once and re-paid on no query. This gives the instrument a second use. An automated extraction candidate (for example a block proposed by OntoCast [14], analysed here purely as a third-party upstream producer) can be scored by the copy ceiling of the questions its block is meant to answer: a candidate whose block drives the ceiling toward 1 is answer-complete and worth promoting; one that does not is not yet ready. The same quantity that audits the serving layer thus grades the ingestion layer—answer-completeness of a block $\approx$ its copy ceiling—turning the control into a governance gate at the boundary where value is actually created.

The honest discovery arc. We did not set out to write this. We built the serving node first, saw large uplift, and worried we had merely reinvented GraphRAG with a nicer provenance story. The copy ceiling began as an adversarial question we put to our own result—*is any of this reasoning, or is it all exposure?*—and the honest answer, uniformly negative gain over copy across ten models and a placebo that collapses once verified inert, is that it is exposure. For a private-knowledge product that is not a disappointment but the specification met: trustworthy because the source is vetted, attributable because the generation is named, and cheap because the curation is paid once. The contribution is the instrument that let us tell the difference, and the discipline of reporting it.

9 The Multivariate Bar

No single number is the deployable target. The bar introduced in §2 is conjunctive across four axes; Table 7 states each, the instrument that measures it, and this work’s honest status against it. We

Table 7: The multivariate bar and this work’s status. A deployable private-knowledge system must clear all four *together*; the axes are measured on different instruments, so no single number establishes the conjunction.

Axis	Instrument & evidence	Status
1. In-domain delivery fidelity	Copy ceiling / gain over copy: ten models -0.067 to -0.022 ; live loom—raw $+0.27$ pooled, $+0.79$ arcane	Measured (v1 + live)
2. Out-of-domain non-regression	Judged Δ vs frontier gold: v1 all CIs include zero (worst -0.23 $[-0.63, +0.05]$); live general $+0.05$ $[0.00, +0.13]$ inside ± 0.25 TOST margin	Measured (v1 + live)
3. Beats web search in-domain	Same model grounded in live web results vs curated scaffold, identical question set	Specified; not measured
4. Attribution precision	Per-answer generation + confidence already emitted; precision metric to close “faithful” $\rightarrow$ “attributable”	Specified; not measured

shrink rather than inflate the scorecard: two axes are measured here, two are specified and not yet closed.

Axes 1 and 2 are the conjunction this paper establishes: excellent, attributable delivery where the corpus is authoritative, *and* no detected regression where it is not, measured on two different instruments so the pairing is not an artefact of one metric. Axis 3 (the right baseline is a web-grounded arm on the same questions, not the bare model) and axis 4 (a reported attribution-precision figure) are designed and unmeasured; we name them as open rather than imply the bar is cleared. The value of stating the full bar is that it prevents the in-domain number from being read as the whole claim—the failure mode the copy ceiling exists to prevent.

10 Limitations and Threats to Validity

- **Single corpus.** Both studies run on one curated ontology in one broad domain. The copy ceiling and its decomposition are corpus-general in construction, but the magnitudes—a 0.964 ceiling, a $+0.27$ live lift—are properties of this corpus and do not transfer as numbers.
- **Single-family judge.** Study 2’s judge is a single GPT-family model. It is cross-family from the Qwen candidate (avoiding self-preference) but is not a panel; a judge idiosyncrasy would propagate. The gold references mitigate but do not eliminate this.
- **Complete-case control attrition.** The control arms were not retried at the larger budget and carry 36–50% empty attrition at 1536, so every control contrast is complete-case over unequal survivor sets. The arms do not sum arithmetically, several are non-significant with wide intervals (shuffled—raw, masked—raw, true—irrelevant), and the attribution rests on the *pattern* across arms rather than any single contrast.
- **Small per-set n for controls.** Per-set control contrasts run on $n = 8$ –21; the curation-depth gradient (arcane true—raw $+0.95$ significant, thin $+0.08$ ns) is suggestive, not definitively estimated, and the thin-set nulls are as consistent with low power as with no effect.
- **Lexical scorer for the v1 axes.** The ten-model gain-over-copy and in-domain recall use a lexical title matcher that under-counts paraphrase; absolute recall is a lower bound and gain over copy is framed as a lower-bound estimand. Only Study 2 uses the semantic judge.
- **Semantic fallback disabled throughout.** The HNSW semantic retrieval path was off in every reported run—its document-embedding recall (0.816) is below the 0.87 design floor (a red gate we report rather than relax)—so the loom arm is lexical-retrieval only. A query-shaped embedding that clears the floor could change the engagement and thin-set numbers.
- **Single production model.** Study 2 holds the model at Qwen3.8-27B. The serving-path effect is isolated cleanly, but its magnitude under other backends is unmeasured; the ten-model sweep

varies the model only for the offline scaffold, not the live node.

- **Three excluded questions.** Three questions empty at 4096 in both arms are dropped (complete-case $n = 114$). The exclusion is symmetric across arms but is a departure from intention-to-treat.
- **Two axes of the bar unmeasured.** The web-search baseline (axis 3) and attribution precision (axis 4) are specified and not measured here; the deployable claim is not yet fully closed.

11 Conclusion

Grounding evaluations whose gold is derived from the injected corpus should **report a copy ceiling as standard**. It is a cheap, per-item, model-free quantity—the recall a verbatim copy of the shown context already achieves—and the signed gain over copy it defines separates faithful delivery of exposed facts from reasoning over injected structure, a distinction a bare-model baseline cannot make. Across ten models the gain is uniformly negative; a production paired study lifts judged quality by +0.27 pooled and +0.79 where curation is deep; and a negative-control decomposition binds that lift to scaffold *content*, with the placebo collapsing once verified seed-disjoint. The reusable method is the pair: *the ceiling plus a verified-inert placebo*, computed before architecting to tell a serving regime (verbatim retrieval wins) from a synthesis regime (where heavier machinery must be measured above the ceiling, not above the bare model). And the economics follow the measurement: value concentrates upstream at curation, where the same instrument doubles as a promotion gate for extraction candidates. For a private corpus the goal was never a model that reasons its way to answers it never saw—it is a model that delivers vetted knowledge faithfully, attributably and cheaply, and an instrument honest enough to prove that is what it does.

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